

PERSONAL DATA SCIENCE PROJECT · 2026 REVISION

# Ten Years of Chicago's Carbon Ledger

Revisiting a 2023 team capstone with fresh data, a rigorously tested question, and Claude Code as a working collaborator.

Harshi Vetrivel · Originally CIF Scientists, Spring 2023 · [github.com/zeal-harshi-vetrivel/chicago-ghg-emissions](https://github.com/zeal-harshi-vetrivel/chicago-ghg-emissions)



# From Team Capstone to Personal Case Study

## 2023 — ORIGINAL

- 7 collaborators, one semester
- 2014-2020 snapshot, 17,728 rows
- A t-test that was defined but never run
- A statistically invalid zip-code correlation
- Regression, a classifier, and a map with no shared throughline

## 2026 — THIS REVISION

- Solo revision, built with Claude Code
- 2014-2023 refresh, 28,329 rows
- Every bug found by actually running the notebook
- Two panel-based questions, tested properly
- Every chart and claim verified before shipping

## Why Revisit This Now?

In September 2026, the EPA repealed its rule limiting greenhouse-gas emissions from power plants — rolling back federal climate policy by roughly a decade.

City ordinances like Chicago's building benchmarking law are increasingly one of the few levers left. This analysis asks what the emissions-intensity trend actually looks like among the buildings it covers - not whether the ordinance caused it.

# A Decade of Data, Not Six Years

Original snapshot

**17,728**

rows · 2014-2020



Refreshed Sept 2026

**28,329**

rows · 2014-2023

Same source, three more years of city reporting, +10,601 rows.



# What Actually Running the Code Found

| Bug                                      | Why it mattered                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Numbers like "104,849" loaded as text    | Every downstream calculation on floor area, electricity use, and GHG emissions would have silently run on strings, not numbers |
| geopandas import commented out           | The map cells later in the notebook would crash with NameError on a fresh run                                                  |
| Row-by-row loops with chained indexing   | Slow, warning-prone cleaning of neighborhoods, spelling fixes, and regions                                                     |
| Zip code correlation and an unrun t-test | Both original "Hypothesis Testing" results were statistically invalid or simply incomplete                                     |



## Two Questions, Tested Properly

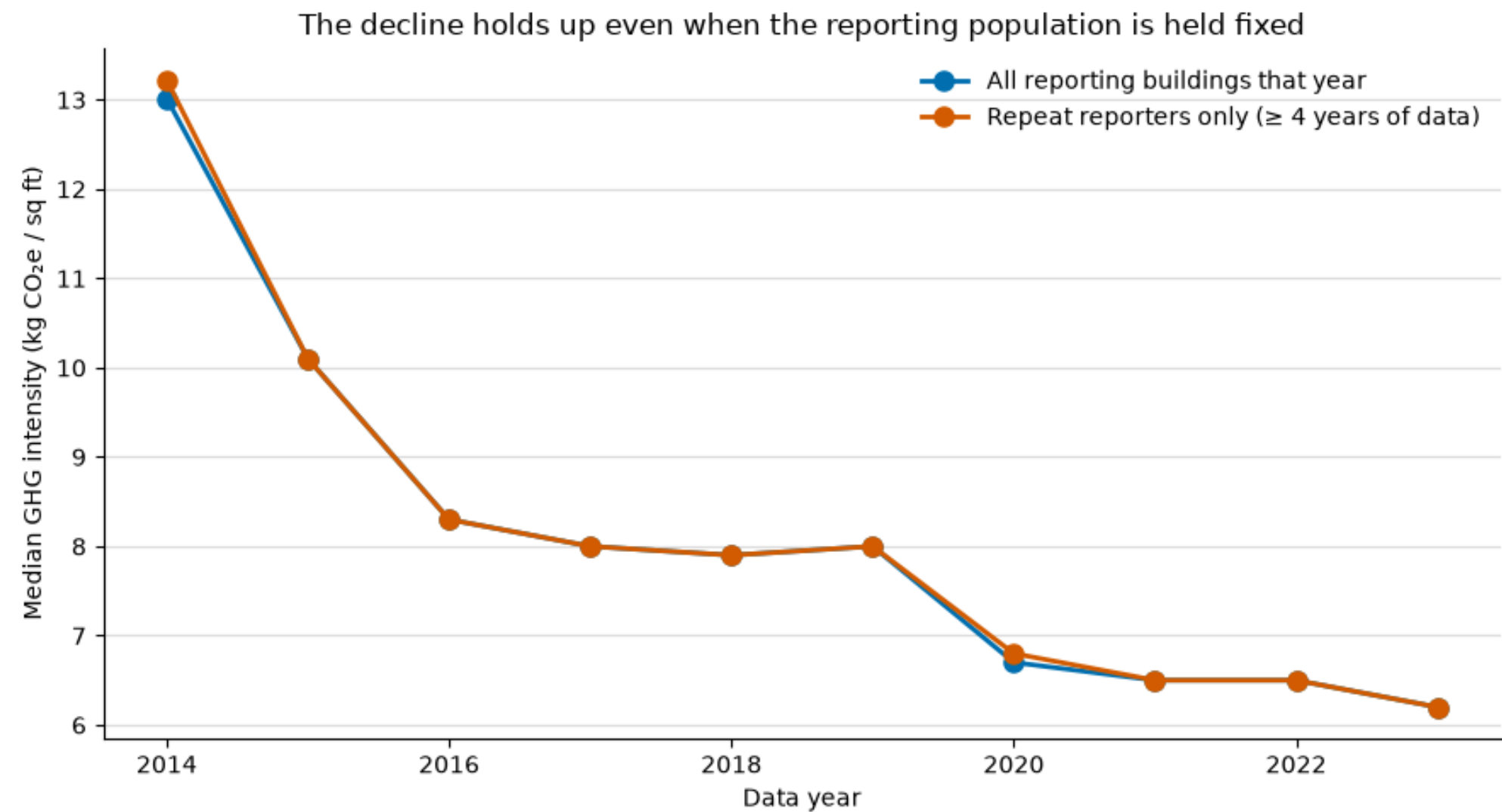
1

Has GHG intensity among buildings in the benchmarking dataset actually declined — or does that just reflect which buildings happen to report each year?

2

Among buildings tracked for years, are they individually cutting their own emissions intensity?

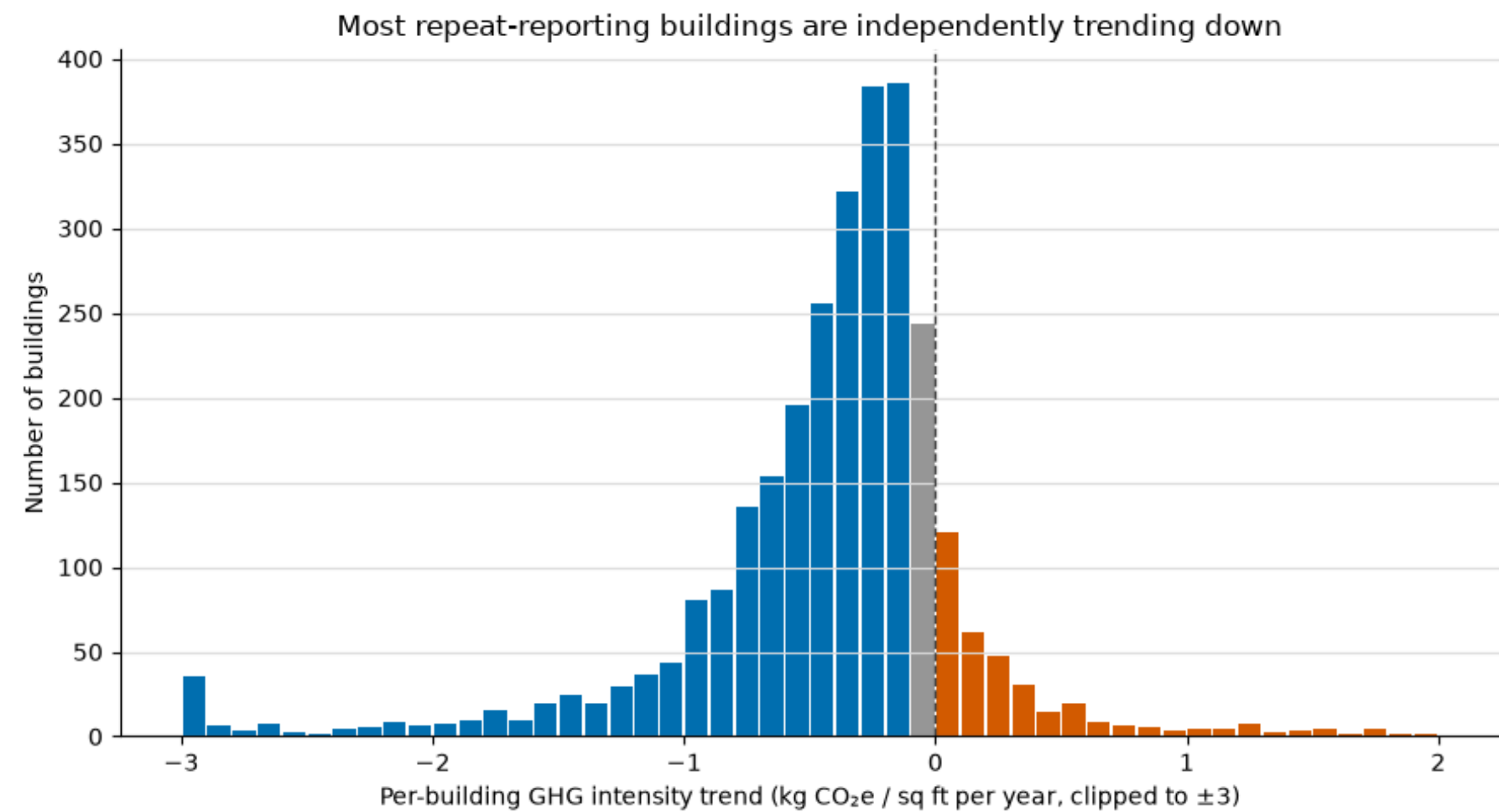
# Finding 1: The Decline Holds Up



Median GHG intensity, all reporters vs. a fixed panel of repeat reporters (4+ years of data) — nearly identical.



# Finding 2: Most Buildings Are Improving on Their Own



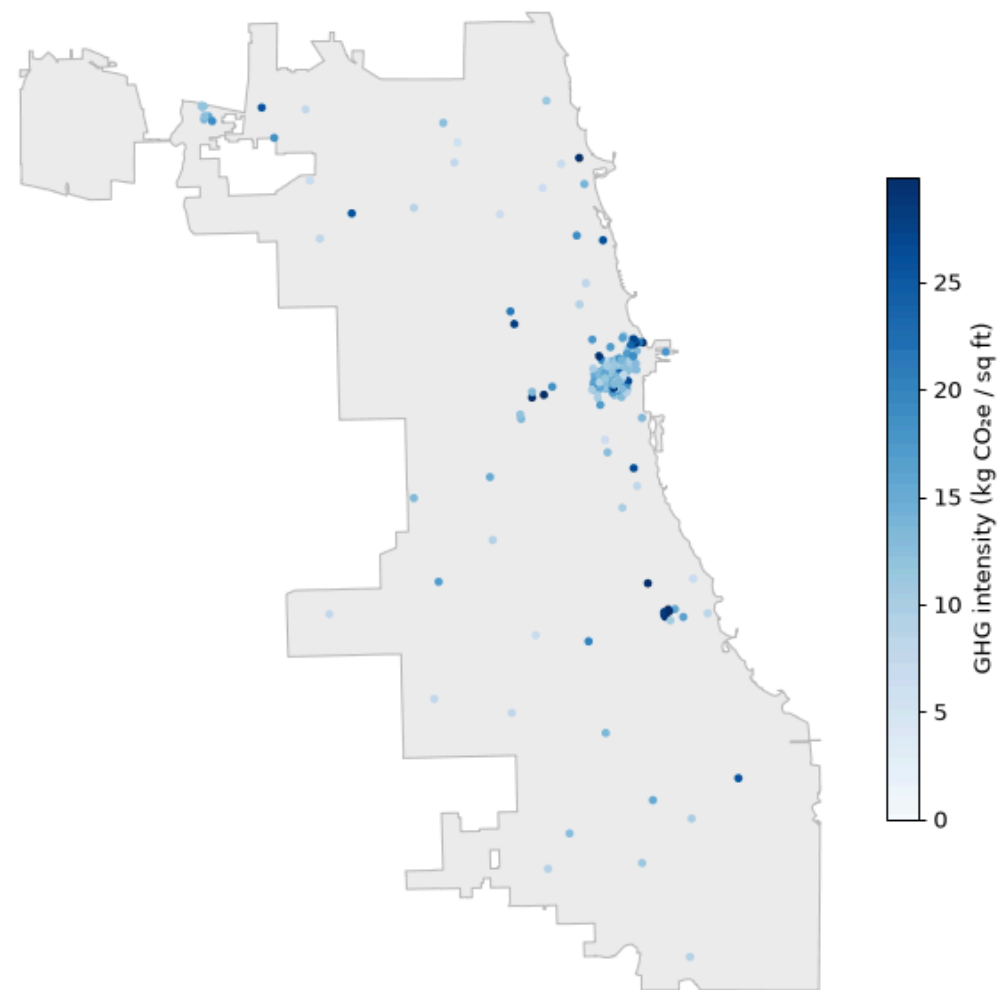
**83.9%**  
of repeat-reporting buildings improved

**9.6%**  
worsened

**6.6%**  
roughly flat

# Same Story, Mapped

GHG Intensity by Building — 2014 (240 reporting)



Two signs of the same decline, visible at once: the point cloud fills in, and the dots trend lighter.

**240 → 2,572**

buildings reporting, 2014 vs. 2023

Reuses `chicago_outline.shp`, a shapefile that sat unused in the repo after the original map section was cut.

# The Trend Is Real — But It's Not That Simple

**–36%**

2014 → 2016

May be inflated by 2014's record-cold winter

**–3.6%**

2016 → 2019

Three years of essentially flat progress

**–16%**

2019 → 2020

A one-year cliff — pandemic vacancy, not retrofits

**–7.5%**

2020 → 2023

Slow, real, but modest since

Federal power-plant rules were repealed in 2026, partly on the premise that cities can carry more of the climate burden. Buildings in this dataset show a real decline in GHG intensity - but this analysis has no comparison group, so it can't isolate the ordinance's own effect from other trends, and the earliest, largest drop coincides with an unusually cold 2014 winter.

**The trend is real. Whether the ordinance caused it is still an open question.**

# How Claude Code Helped — and What I Directed

## I SET THE DIRECTION

- Chose which research questions were worth asking
- Decided what to prune versus keep, for one clear throughline
- Caught that the panel barely differed from the raw average — and that mattered
- Asked for a map that showed the trend, not just decoration

## CLAUDE CODE EXECUTED & VERIFIED

- Found the thousands-separator and missing-import bugs by running the code, not just reading it
- Re-ran the full notebook end to end after every single change
- Applied a colorblind-safe, single-axis chart design system
- Was transparent about a network block instead of fabricating data



# A Real Exchange From This Project

**Me:**

"What if we make a map visual that does show the trend across time — like a video?"

**Claude Code:**

Built an animated map reusing the shapefile that had sat unused since the original project — one frame per year, colored by GHG intensity, verified by executing the notebook and reviewing every frame before shipping it.



# Skills This Project Demonstrates

- Data Cleaning & Validation

- Reproducible Analysis

- Panel & Composition-Bias Reasoning

- AI-Assisted Engineering

- Data Visualization Design

- Git & Version Control

# Explore the Full Analysis

[github.com/zeal-harshi-vetrivel/chicago-ghg-emissions](https://github.com/zeal-harshi-vetrivel/chicago-ghg-emissions)

Data cleaning, a panel-based trend analysis, an animated map, and a full commit history showing how this project — and the way I use AI tools — has grown.